Arista Access Network Hardware Failure Resolution: WFD-ACC-48 - Whitefield POP

Node: WFD-ACC-48 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_9155

Description

The WFD-ACC-48 access network device has suffered a hardware/environmental fault resulting in the loss of power supply redundancy and the subsequent crash of the line card, manifesting as ASIC memory parity errors. This failure domain is inherent to the Arista access network architecture, where the line card serves as the primary entry point for incoming traffic. In the event of a line card crash, the ASIC memory parity errors trigger the alarm ALARM_9155, necessitating immediate intervention to prevent service disruption. The root cause of this failure lies in the hardware design, where the loss of power supply redundancy compromises the stability of the line card and, subsequently, the ASIC memory.

Pre-requisite Checks

- Check: Verify the power supply redundancy status on the Whitefield POP site, Status: Confirmed, Action: Dispatch remote hands to Whitefield POP to verify power supply redundancy status

- Check: Check the redundant link status, Status: Confirmed, Action: Verify the status of the redundant link using the 'show interfaces' command

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crash	Medium
Loss of power supply redundancy	Low

Diagnostic Commands & Expected Outputs

```
Command: show system internal switch, Expected Output: System Internal Switch Information, Description: This command displays the system internal switch information, including the ASIC memory parity errors, which are indicative of the hardware failure.

Command: show interfaces, Expected Output: Interface Information, Description: This command displays the interface information, including the status of the redundant link, which may be affected by the hardware failure.

Command: show log, Expected Output: System Log Information, Description: This command displays the system log information, including any error messages related to the hardware failure.
```

Resolution Steps

Step 1: Step: Replace the line card, Command: replace line-card , Description: Replace the faulty line card with a new one to restore the functionality of the access network device.

Step 2: Step: Verify the power supply redundancy status, Command: show power supply, Description: Verify the power supply redundancy status to ensure that it is restored to a stable state.

Step 3: Step: Verify the redundant link status, Command: show interfaces, Description: Verify the status of the redundant link to ensure that it is functioning correctly.

Post-Check Verification

```
Command: show system internal switch, Expected Output: System Internal Switch Information, Description: Verify that the ASIC memory parity errors are resolved.

Command: show interfaces, Expected Output: Interface Information, Description: Verify that the redundant link is functioning correctly.

Command: show log, Expected Output: System Log Information, Description: Verify that there are no error messages related to the hardware failure.
```

Roll-back Procedures

- Step: Restore the original line card, Command: replace line-card revert, Description: Restore the original line card if the replacement fails.

- Step: Re-verify the power supply redundancy status, Command: show power supply, Description: Re-verify the power supply redundancy status to ensure that it is restored to a stable state.

- Step: Re-verify the redundant link status, Command: show interfaces, Description: Re-verify the status of the redundant link to ensure that it is functioning correctly.